X-ray Cavities Honours Project: Stewart Environment Gate Status

Supervisor update for 2026-05-06 meeting

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2026-05-05

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Executive Summary

The main science asset blocker has been cleared. Patrick supplied the Stewart environment archive and GRAiLE/XRAiSE bundle, and Stas clarified the key science choice: production science runs should use the radially averaged tabulated environments, not the full 3D grid environments, because the full 3D profiles introduce too much spurious motion for reliable energetics.

We extracted the two selected tabulated Stewart environments, staged them to a resized Nectar VM, and ran no-jet 3D Cartesian stability gates to `300 Myr`. Both the cluster and group environments passed the existing `jet-setup` stability verifier at smoke scale.

The project is now ready for the next science gate: a particle-free `C-HI` high-power cluster rehearsal on resized Nectar, followed by a particle-enabled repeat only after the hydrodynamic run is stable and diagnostically useful.

What Is Now Confirmed

Environment Selection

Case role	Case ID	Stewart environment	Selected file
high-power cluster validation	`C-HI`	cluster `002_003`	`tabulated_cosmo_002003.csv`
low-power cluster science anchor	`C-LO`	cluster `002_003`	`tabulated_cosmo_002003.csv`
low-power group comparison	`G-LO`	group `002_031`	`tabulated_cosmo_002031.csv`

The full 3D environment cubes remain useful reference assets, but are not the default science inputs for the honours energetics path.

Frozen Science Defaults

Parameter	Current default
Physics	HD
Geometry	3D Cartesian
Jet speed	`0.1 c`
Opening angle	`15 deg`
Duration	`300 Myr`
Snapshots	`100`, `200`, `300 Myr`
Views	`0`, `45`, `90 deg`
First validation case	`C-HI`, `JET_PWR = 1e45 erg/s`
Particle staging	off first, on second

Important unit correction: the old Stewart INIs confirmed that PLUTO time values are in code-time units. A `300 Myr` run is rendered as:

```
tstop = 91980.41813640
JET_END_TIME = 91980.41813640
JET_ACTIVE_TIME = 91980.41813640
100 Myr output cadence = 30660.13937880
```

Nectar Environment Gate

The MacBook remained the control plane. `nectar-vm` was resized to `c3.3xlarge` for the gate tests, then parked back to `c3.xsmall` afterwards.

Each gate used:

- HD, 3D Cartesian, vector body force
- no jet: `JET_END_TIME = 0`, `JET_ACTIVE_TIME = 0`
- no particles: `PARTICLES = NO`
- uniform `50 x 50 x 50` grid over `[-100, 100] kpc`
- outputs near `0`, `100`, `200`, and `300 Myr`

Results

Environment	Runtime	Max central density drift	Max central pressure drift	Max Mach	Max velocity	Gate
cluster 002_003	39 s	0.136%	0.226%	0.0242	10.6 km/s	pass
group 002_031	30 s	0.284%	0.473%	0.0330	8.07 km/s	pass

The result is not a final production-resolution validation, but it is a clean smoke-scale gate: both tabulated environments load correctly, remain quiet over the full `300 Myr` duration, and produce standard `grid.out`, `dbl.out`, and `data.*.dbl` outputs that load through the current Python analysis environment.

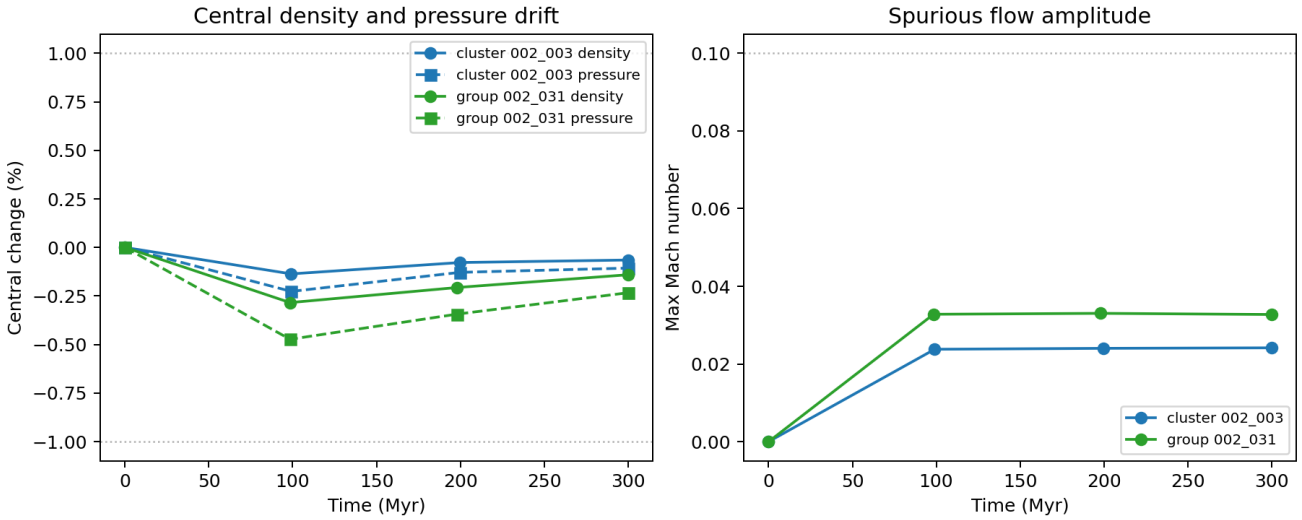


Figure 1: Stability gate time series

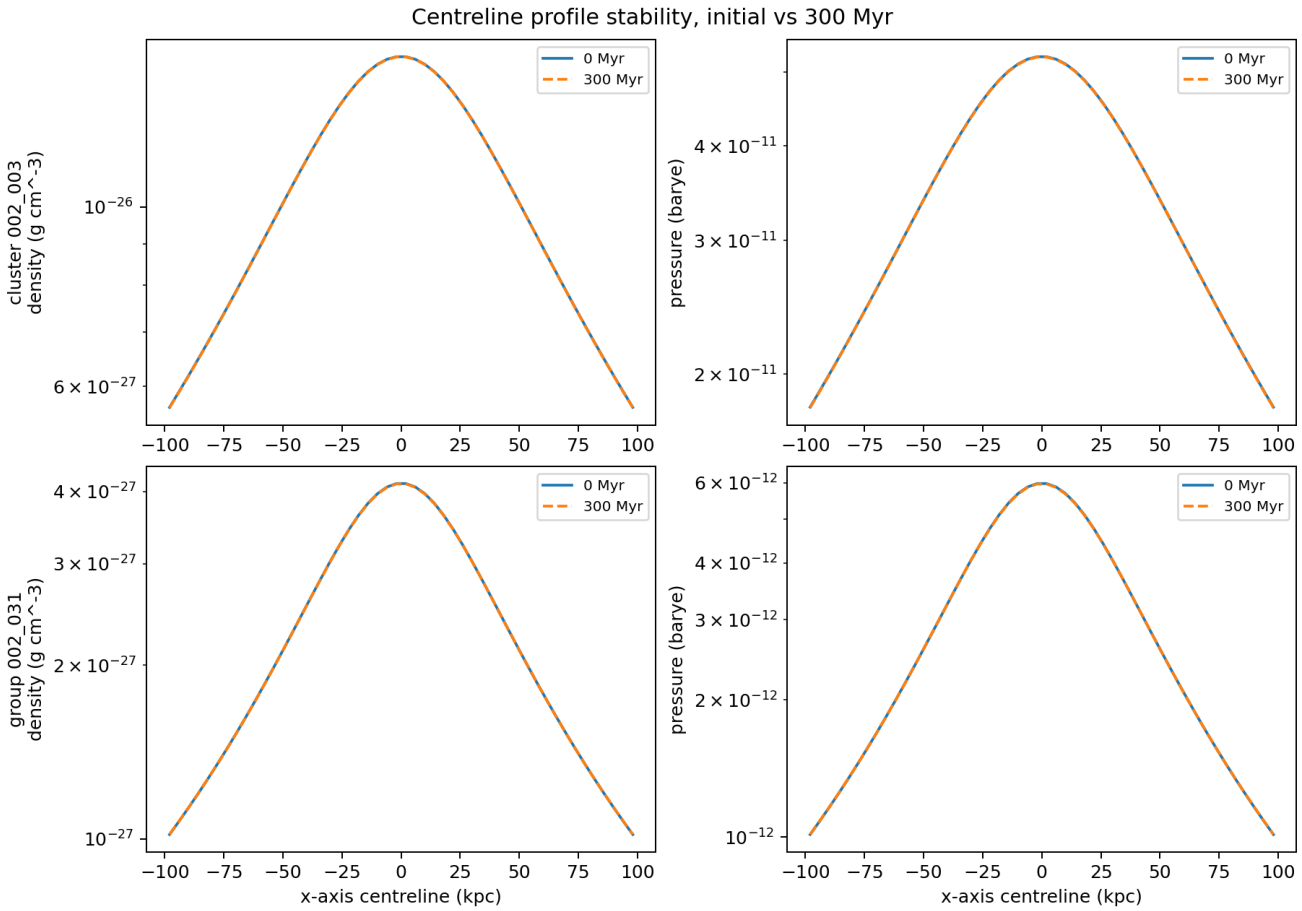


Figure 2: Initial versus 300 Myr centreline profiles

Technical Work Completed

New and Updated Project Files

- `docs/science/stewart_environment_gate_2026-05-05.md`: detailed gate report.
- `docs/science/assets/stewart_environment_gate_2026-05-05/`: metrics and plots.
- `scripts/science/render_stewart_environment_gate_tests.py`: renders `jet-setup`-compatible no-jet stability tests from the Stewart tabulated environments.
- `scripts/science/plot_stewart_environment_gate.py`: regenerates the summary metrics and plots from gate outputs.
- `scripts/science/render_three_case_configs.py`: now renders jet active/end times in PLUTO code-time units.
- `docs/science/three_case_spec.md`: updated to record the tabulated environment decision, the green Nectar gate, and the code-time convention.

Operational Fixes

The current `jet-setup` source template keeps `PARTICLES_LP` enabled in `definitions.h`. For a true no-particle environment gate, the gate renderer now emits a hydro-only `definitions.h` and a `local_make`

that includes the UTAS environment and jet user objects without including `Particles/makefile`.

Nectar SSH was also made more stable after intermittent port 22 connection refusals by disabling socket activation and enabling persistent `ssh.service`. After the gate, `nectar-vm` was parked back to `c3.xsmall` and verified by SSH as `1` vCPU and about `1.9 GiB` RAM.

Current Status

The infrastructure path is green:

- MacBook can orchestrate Nectar.
- Nectar can resize up for rehearsals and park down afterwards.
- `kunanyi` has already passed PBS, HDF5, PLUTO, `jet-setup`, email, and multi-node MPI smoke tests.

The science-input path is now green for the first rehearsal:

- Stewart tabulated cluster and group files are identified.
- Both selected files pass no-jet stability smoke tests.
- The GRAiLE/XRAiSE bundle is present locally, but still needs path and dependency porting before it is a repo-native X-ray pipeline.

Immediate Forward Plan

Gate 1: `C-HI` Hydrodynamic Rehearsal on Nectar

Run the first high-power cluster case with particles off:

- case: `C-HI`
- environment: cluster `002_003` tabulated profile
- jet power: `1e45 erg/s`
- jet speed: `0.1 c`
- opening angle: `15 deg`
- duration: `300 Myr`
- outputs: sparse `dbl` at `100`, `200`, `300 Myr`, plus HDF5/checkpoint hooks

Pass criteria:

- stable completion to `300 Myr`
- clear lobe/cavity morphology
- density, pressure, tracer, and log diagnostics produced
- no output, restart, or code-time surprises

Gate 2: Particle-enabled `C-HI` Repeat

Repeat the same case with particles enabled only after Gate 1 is green.

Pass criteria:

- particle output files are produced and load
- `plutokore` radio functions can produce at least one smoke product
- radio/tracer/lobe products align at one declared snapshot and view

Gate 3: First Observable/Truth Comparison

Build the first aligned comparison package for one `C-HI` snapshot/view:

- Level 1 truth: injected energy, with one-sided and bipolar conventions both reported until supervisors choose the convention.
- Level 2 truth: tracer-defined lobe volume and energy with tracer threshold `5e-3`, plus sensitivity checks at `1e-3` and `1e-2`.
- X-ray path: port GRAiLE/XRAiSE scripts rather than inventing a substitute.
- Cavity protocol: ellipsoidal cavity boundary, local ambient pressure, `pV` and `4pV`, and sound/buoyancy/refill age estimates.

Gate 4: `kunanyi` Intermediate Rehearsal

Only after the Nectar `C-HI` hydro and particle gates are green should we move to an intermediate `kunanyi` rehearsal using the production PBS template:

```
select=N:ncpus=28:mpiprocs=28
rank count from PBS_NODEFILE
#PBS -M barry.brook@utas.edu.au
#PBS -m ae
```

The production launch order remains:

1. `C-HI`
2. `C-LO`
3. `G-LO`

Supervisor Checkpoints

The useful discussion points for the 2026-05-06 meeting are:

1. Confirm that the tabulated Stewart files are now the agreed science inputs.
2. Confirm that `C-HI` remains the first rehearsal case.
3. Confirm whether `JET_PWR` should be treated as one-sided or total bipolar power in final comparison tables.
4. Confirm whether `JET_INITIAL_RADIUS` and `JET_INJECTION_HEIGHT` should stay near the historical Stewart values or be adjusted for the honours grids.

5. Confirm the preferred first snapshot/view for the aligned X-ray/radio/truth smoke comparison, if not `100 Myr` and `45 deg`.

Bottom Line

We are no longer blocked on the Stewart environment files. The next meaningful science task is a live `C-HI` hydrodynamic rehearsal on resized Nectar, with the particle and observable/truth paths staged behind it.